

DevOps AWS Assignment - Final Report

Name: [Your Name]

Date: [Submission Date]

GitHub repo: [link]

EC2 public IP / demo link: [link]

1. What I built

I deployed a simple Node.js (Express) app on an AWS EC2 instance. The app has a couple of API routes and a small homepage. I set up a GitHub Actions pipeline that deploys the app automatically when I push to the main branch, added basic monitoring with CloudWatch, and ran a load test using k6.

2. Steps I followed

- Launched a t2.micro EC2 instance (Free Tier) on Amazon Linux.
- Installed Node.js, npm, and PM2 on the instance.
- Cloned my GitHub repo and started the app using PM2.
- Created a security group allowing SSH (my IP only), HTTP, and the app port.
- Created an S3 bucket for storing backups/logs.
- Created an IAM role with limited permissions and attached it to the EC2 instance.
- Set up a GitHub Actions workflow to deploy automatically on push.
- Installed the CloudWatch agent and created a basic dashboard + alarm.
- Ran a load test using k6.

3. Monitoring

I created a CloudWatch dashboard to see CPU usage and set up an alarm to notify me if CPU usage crosses 80%. Screenshots are attached separately.

[Insert CloudWatch dashboard screenshot here]

4. Load Testing Results

I used k6 with 20 virtual users for 30 seconds against the /api/items endpoint.

- Average response time: [fill in]
- Requests per second: [fill in]
- Error rate: [fill in]
- CPU usage during test (from CloudWatch): [fill in]

[Insert k6 output screenshot / graph here]

5. Problems I faced

- [Example: had trouble connecting to EC2 at first because of a security group rule]
- [Example: GitHub Actions SSH step failed because of key formatting]

[Add your own real problems and how you solved them]

6. What I would do differently next time

- Set up HTTPS properly with a real domain.
- Use SSM instead of SSH so I do not need to open port 22.
- Write a few automated tests before deploying.
- Try auto scaling if I had more time.

7. Conclusion

This assignment helped me understand how to take a simple app from local development to a live deployment on AWS, with monitoring and a basic CI/CD pipeline. There is still a lot I want to learn about scaling and security, and I have listed some of that above.